FORM FOUR PRE-NATIONAL EXAMINATION

036 INFORMATION AND COMPUTER STUDIES
MARKING SCHEME 2026 – MAJIBU KWA MANENO RAHISI

Nimefuata majibu yaliyopo kwenye PDF ya awali, kuanzia Section A hadi Section C. Sijabadilisha majibu, mpangilio, idadi ya points, wala maana ya content — nimebadilisha tu maneno magumu kuwa maneno rahisi ya kueleweka na mwanafunzi wa Form Four.

SECTION A – 16 MARKS

1. Multiple Choice Answers

Question	Answer
(i)	B
(ii)	C
(iii)	D
(iv)	A
(v)	C
(vi)	E
(vii)	C
(viii)	A
(ix)	B
(x)	B

2. Matching Items

Question	Answer
(i)	C
(ii)	B
(iii)	A
(iv)	D
(v)	E
(vi)	F

SECTION B – 54 MARKS

3. Malware

(a) Three Warning Signs of Malware Infection

• The computer suddenly becomes slow and takes a long time to start.

• Many pop-up windows appear without permission, strange error messages appear, or files change into shortcuts.

• Data files suddenly disappear, file names or extensions change unexpectedly, or system files are deleted.

(b) Three Common Entry Vectors

• Connecting infected flash drives or other external storage devices.

• Downloading files or attachments from unknown emails or websites.
• Using local networks that do not have updated security firewalls.

(c) Three Preventive Measures

• Install good antivirus and anti-malware software and update them regularly.
• Keep the operating system security updates and network firewalls updated.
• Give users regular security training about safe email use and avoiding illegal software downloads.

4. Information Systems

(a) Three Reasons for Protecting Information Systems

Any three:

• Prevent people without permission from accessing information.
• Keep information private.
• Keep data correct and unchanged.
• Prevent loss of money.
• Make sure the business continues to operate.
• Increase customer trust.

(b) Three ICT-related Crimes

Examples:

• Hacking
• Stealing someone's identity
• Phishing
• Illegal copying of software
• Online fraud
• Spreading harmful software
• Cyberbullying
• Online money fraud

(c) Security Measures

• Install antivirus software.
• Use strong passwords.
• Protect sensitive data by changing it into a secure form (encryption).
• Set up firewalls.
• Update software regularly.
• Control what each user is allowed to access.
• Teach users about cyber security.

5. Health and Ergonomics

(a) Three Health Risks

• Pain or injury in the fingers and wrists caused by using the computer repeatedly for a long time.
• Computer Vision Syndrome, which can cause dry eyes, unclear vision and headaches.
• Long-term pain in the neck, shoulders and lower back caused by sitting in a poor position.

(b) Three Ergonomic Solutions

• Adjust the chair height and back support so that your back stays in a good position and your feet are flat on the floor.

• Put the computer screen at eye level and about one arm's length away to reduce neck pain and eye tiredness.

• Use a comfortable keyboard and wrist support to keep your wrists straight and in a natural position when typing.

6. Electronic Approach and Cloud Computing

(a) Two Advantages of Electronic Approach

Faster Processing and Entry: Data is sent directly to a central database. This reduces mistakes caused by entering information on paper and reduces delays.

Built-in Checking Rules: The software checks wrong or unusual information immediately when it is entered, which improves data accuracy.

(b) Three Field Challenges

• Sudden system failures or the battery running out in areas where electricity is not available.

• Poor or unstable mobile network connections, which stop data from being sent immediately.

• Hardware problems, such as broken screens or damage caused by water or other difficult outdoor conditions.

(c) Two Advantages of Cloud Computing

Any two:

• It allows data to be updated and shared immediately.

• It allows people to access information from different places.

• It reduces the risk of losing data by keeping backup copies.

• It makes sharing information easier.

• It reduces the need to store everything on a local computer.

7. Microsoft Excel

(a) Steps for Conditional Formatting

• Select the cells you want to work on using the mouse.

• Click the Home tab, find the Styles group, and open Conditional Formatting.

• Select Highlight Cells Rules → Less Than, type 50 in the value box, choose Red Fill, and click OK.

(b) Steps for Freezing the Top Row

• Open the worksheet and click the View tab on the main menu.

• Find the Window group and click the Freeze Panes button.

• Select Freeze Top Row to keep the first row visible.

(c) Steps for Protecting a Worksheet

• Click the Review tab on the main toolbar.

• Find the Protect group and click Protect Sheet.

• Enter a secure password, choose what users are allowed to do, click OK, enter the password again to confirm, and save the file.

8. Computer Storage and Memory

(a) Storage Capacity and Processor Speed

• Hard disk storage capacity: 1 TB

• Processor speed: 3.2 GHz

(b) Functions of Hard Disk

Any two:

- Stores programmes permanently.
- Stores users' files.
- Stores the operating system.
- Stores application software.
- Stores backup copies.

(c) Functions of RAM

- Temporarily stores data that is being processed.
- Temporarily stores programmes that are running.
- Gives the CPU data quickly.
- Keeps temporary results while the computer is processing data.

SECTION C – 30 MARKS

9. Multimedia Technology

Introduction

Multimedia technology is the use of different types of media together, such as text, pictures, sound, animation and video, using digital technology. It is growing and helping economic development and modern communication in Tanzania.

Main Body

Education Sector: Interactive animations and digital video lessons make difficult STEM subjects easier for students to understand in schools. E-learning platforms also allow students to access learning materials even when they are outside the classroom.

Marketing and Business: Businesses use attractive digital video advertisements and pictures on social media to make more people know about their brands and increase sales. This helps businesses grow through digital technology.

Entertainment and Creative Arts: Local film and music industries use modern digital tools for editing sound and video to produce high-quality media. This creates jobs for young people and increases money earned from entertainment in the country.

Public Health and Civic Campaigns: Government organisations use multimedia pictures and radio/TV animations to share important health information, such as vaccination campaigns and disease prevention. This increases public awareness and safety.

Conclusion

Multimedia technology has changed traditional ways of communication. Continued investment in digital skills training and internet infrastructure is important to get the full social and economic benefits of multimedia technology.

10. Operating System

Introduction

An Operating System (OS) is the main software that controls computer hardware and provides common services for application programmes. Choosing the right operating system requires considering the computer hardware and the needs of the institution.

Main Body

Hardware Compatibility and System Requirements: Check whether the existing processor, RAM and storage space meet the minimum requirements needed for the operating system to work properly.

Built-in Security: Check the security features included in the operating system, such as firewalls, user access controls and automatic data protection, to protect against data theft.

Total Licensing and Maintenance Costs: Consider the cost of buying the software licence, free or open-source options, and the cost of technical support in the future.

User Interface Design and Ease of Use: Check whether the Graphical User Interface (GUI) is easy to use so that staff can use the system properly without needing a lot of extra training.

Application Software Compatibility: Make sure the new operating system can support the software, databases and drivers needed for the institution's daily work.

Availability of System Updates and Patches: Make sure the software company provides regular and reliable security updates and fixes to protect the computer from new digital threats.

Conclusion

Choosing an operating system should not be based only on how popular the brand is. A proper technical check helps IT managers choose an operating system that provides stability, security and good performance for a long time.

11. Enterprise Resource Planning (ERP)

Introduction

ERP (Enterprise Resource Planning) is a system that brings together and manages major activities of an organisation using one common database.

Main Body

Any six:

- Brings information from all departments together.
- Helps people make better decisions by providing current information.
- Reduces repeated data.
- Improves communication between departments.
- Increases employee productivity.
- Improves management of stock.
- Improves accuracy of payments.
- Improves customer service.
- Helps manage the organisation's money better.
- Reduces operating costs.
- Improves preparation of reports.
- Improves data security through central management.

Conclusion

ERP improves the efficiency of an organisation by bringing business activities together, reducing costs, improving decision-making and increasing productivity.